

StarForth Formal Verification Report

Isabelle/HOL Machine-Checked Proofs of Determinism and Correctness

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Part I

Foundations

Chapter 1

Verification Methodology

Chapter 2

Base Definitions and Type System

Part II

The Seven Feedback Loops

Chapter 3

Loop 1 — Execution Heat

Chapter 4

Loop 2 — Rolling Window of Truth

Chapter 5

Loop 3 — Linear Decay

Chapter 6

Loop 4 — Pipelining and Prefetch

Chapter 7

Loop 5 — Window Width Inference

Chapter 8

Loop 6 — Decay Slope Inference

Chapter 9

Loop 7 — Adaptive Heartrate

Part III

Word Category Correctness

Chapter 10

Arithmetic Words

Chapter 11

Stack Words

Chapter 12

Logical Words

Chapter 13

Memory Words

Chapter 14

Return Stack Words

Chapter 15

Q48.16 Fixed-Point Arithmetic

Part IV

System-Level Proofs

Chapter 16

Overall Correctness

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State Transitions

Chapter 18

Concurrency Properties

Chapter 19

Mutual Exclusion

Part V

Word-Level ACL Proofs

Chapter 20

Pin Monotonicity

Chapter 21

Inheritance Clears Pin

Chapter 22

TTL Boundedness

Chapter 23

Emergency Bypass

Chapter 24

No Privilege Escalation

Appendix A

Theory File Inventory

Table A.1: Isabelle/HOL theory files (24 total)

Theory file	Coverage
StarForth_Base.thy	Base definitions, type system
StarForth_Loop1_Heat.thy	Execution heat (Loop 1)
StarForth_Loop2_Window.thy	Rolling window (Loop 2)
StarForth_Loop3_Decay.thy	Linear decay (Loop 3)
StarForth_Loop4_Pipeline.thy	Pipelining (Loop 4)
StarForth_Loop5_WinInf.thy	Window inference (Loop 5)
StarForth_Loop6_DecayInf.thy	Decay inference (Loop 6)
StarForth_Loop7_Heartrate.thy	Adaptive heartrate (Loop 7)
StarForth_Arithmetic_Words.thy	Arithmetic word correctness
StarForth_Stack_Words.thy	Stack word correctness
StarForth_Logical_Words.thy	Logical word correctness
StarForth_Memory_Words.thy	Memory word correctness
StarForth_Return_Stack_Words.thy	Return stack correctness
StarForth_Q48_16.thy	Q48.16 fixed-point
StarForth_Correctness.thy	Overall correctness
StarForth_Transition.thy	State transitions
StarForth_Concurrent.thy	Concurrency properties
StarForth_Mutex.thy	Mutual exclusion
ACL_Pin_Monotone.thy	ACL pin monotonicity
ACL_Inherit_Clears_Pin.thy	ACL inheritance
ACL_TTL_Bounded.thy	ACL TTL bounds
ACL_Emergency_Bypass.thy	ACL emergency bypass
ACL_No_Escalation.thy	ACL no escalation
ROOT	Isabelle session manifest

Appendix B

Building and Checking Proofs

